SENLA-ResUNet: A Hybrid Residual U-Net with Squeeze-Excitation and Non-Local Attention for Accurate Nuclei Segmentation

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ABSTRACT

Accurate nuclei segmentation is a fundamental step in quantitative biomedical image analysis, enabling critical applications in disease diagnosis, prognosis, and therapeutic response assessment. Despite advancements in deep learning-based segmentation models, challenges persist due to variations in nuclear morphology, diverse imaging modalities, and intricate cellular environments, often limiting the precision of boundary delineation. This paper introduces SENLA-ResUNet, a novel hybrid deep learning architecture designed for robust and highly accurate nuclei segmentation. Our model builds upon the efficient Residual U-Net framework by strategically integrating advanced attention mechanisms: Squeeze-and-Excitation (SE) blocks for adaptive channel-wise feature recalibration, Non-Local Attention for capturing long-range contextual dependencies, and Inception modules within an encoder attention block to extract multi-scale features. Furthermore, attention gates are incorporated into the skip connections of the SENLA-ResUNet decoder to selectively highlight salient information and refine feature propagation. Evaluated on the challenging Data Science Bowl 2018 (DSB-2018) and Triple Negative Breast Cancer (TNBC) datasets, SENLA-ResUNet consistently demonstrates superior performance compared to established baselines including standard U-Net, ResUnet, Attention U-Net, and Cellpose. Our model achieves an Intersection over Union (IoU) of 0.7936 and a Dice Score of 0.8172, alongside high precision (0.8791), recall (0.9084), and accuracy (0.9726), setting a new benchmark for nuclei segmentation of DSB-2018 dataset. These results underscore the efficacy of combining residual learning with multi-faceted attention mechanisms for enhanced feature representation and precise pixel-level segmentation. Our code is open source and available at https://github.com/OVER-CODER/Nuclei-Segmentation.

Keywords:
Image Segmentation, Medical Imaging, Residual U-Net, Attention Mechanism, Squeeze-and-Excitation, Deep Learning, Nuclei Segmentation.

1. INTRODUCTION

Accurate segmentation of biological structures within microscopy images plays a pivotal role in a wide array of biomedical research and clinical applications. Notably, precise nuclei segmentation is fundamental for quantitative analysis in histopathology, cell biology studies, and drug discovery, enabling tasks such as cell counting, morphometric analysis, and the understanding of tissue architecture [1]. However, achieving robust and accurate nuclei segmentation remains a challenging task due to the inherent complexity of biological images, including variations in cell size, shape, staining intensity, the presence of cellular clumps, and image artifacts [2].

In recent years, deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized the field of medical image analysis, demonstrating remarkable success in various segmentation tasks [3]. The U-Net architecture [4], with its symmetric encoder-decoder structure and crucial skip connections, has become a foundational model for biomedical image segmentation, effectively capturing both high-resolution spatial details and low-resolution contextual information. Subsequent research has focused on enhancing the U-Net architecture to address its limitations and further improve segmentation performance.

One significant area of improvement involves the integration of residual connections, leading to architectures like ResUnet [5]. Residual connections facilitate the training of deeper networks by mitigating the vanishing gradient problem and allowing the network to learn identity mappings, thereby improving information flow and overall performance. Another direction of research explores the incorporation of attention mechanisms into U-Net-like architectures, such as the Attention U-Net [6]. Attention mechanisms enable the network to selectively focus on the most relevant spatial regions and feature channels, suppressing irrelevant noise and improving the accuracy of segmentation, particularly in complex scenarios with indistinct boundaries or cluttered fields.

While these advancements have yielded significant progress, effectively capturing both channel-wise relationships and long-range spatial dependencies across varying scales is crucial for achieving highly accurate and robust nuclei segmentation. Nuclei exhibit diverse sizes and morphologies, and their precise delineation requires a model that can intelligently weight informative features and establish global contextual understanding. Therefore, a segmentation model that can effectively integrate fine-grained local details with broad contextual cues is highly desirable, especially when dealing with densely packed nuclei or subtle visual cues.

To address these challenges and further enhance segmentation precision, this paper proposes:

- **SENLA-ResUNet**, a novel hybrid deep learning architecture for accurate nuclei segmentation. Our approach builds upon the strengths of the Residual U-Net by incorporating advanced attention mechanisms and multi-scale feature extraction directly into its structure.
- SENLA-ResUNet integrates **Squeeze-and-Excitation (SE) blocks** to perform dynamic channel-wise feature recalibration, allowing the model to adaptively emphasize more informative feature channels.
- We introduce **Non-Local Attention** to capture long-range dependencies, enabling the model to learn relationships between distant pixels and enhance global contextual understanding.
- Our architecture incorporates **Inception-like modules** [7] within an encoder attention mechanism at the network's bottleneck to extract robust multi-scale features before applying non-local attention.
- **Attention gates** [6] are strategically placed in the SENLA-ResUNet's skip connections, directing the model to focus on salient regions and filtering out irrelevant information from the high-resolution features transferred from the encoder to the decoder.

The rest of this paper is organized as follows: Section 2 provides a review of related work in medical image segmentation and attention mechanisms. Section 3 details the architecture of the proposed SENLA-ResUNet. Section 4 describes the experimental setup, including the datasets, evaluation metrics, implementation details, and discusses the experimental results, comparing the performance of

SENLA-ResUNet with other methods. Finally, Section 5 concludes the paper and outlines potential directions for future research.

2. Related Work

Accurate and robust nuclei segmentation in medical images has been a subject of extensive research. Traditional approaches often relied on handcrafted features such as shape, size, and texture to distinguish nuclei from the background and other cellular components [8, 9]. While these methods can be effective under specific conditions, they often struggle with the inherent variability in biological samples and the complexity of microscopy images, leading to limitations in generalizability and accuracy across diverse datasets.

The advent of deep learning has significantly advanced the field of medical image segmentation. Convolutional Neural Networks (CNNs) have demonstrated remarkable capabilities in automatically learning hierarchical features directly from image data, leading to substantial improvements in segmentation accuracy [10, 11]. The U-Net architecture [12], with its encoder-decoder structure and skip connections, has become a cornerstone for medical image segmentation tasks, including nuclei segmentation [13, 14]. The encoder path progressively downsamples the input image, capturing contextual information, while the decoder path upsamples the feature maps, enabling precise localization of the structures of interest. Skip connections facilitate the transfer of high-resolution features from the encoder to the decoder, preserving fine-grained details crucial for accurate boundary delineation.

Building upon the U-Net framework, various modifications and enhancements have been proposed to further improve nuclei segmentation performance. One notable advancement is the integration of residual connections, as seen in ResUnet [5]. By introducing identity mappings, residual blocks help to mitigate the vanishing gradient problem, allowing for the training of deeper and more complex networks that can learn more intricate features. This has been shown to improve segmentation accuracy and robustness in various medical image analysis tasks by facilitating better information flow through the network.

Another significant direction in medical image segmentation research involves the incorporation of attention mechanisms. Attention mechanisms enable neural networks to selectively focus on the most relevant parts of the input data, effectively suppressing irrelevant information and improving the model's ability to learn discriminative features. In the context of nuclei segmentation, attention mechanisms can help the network to concentrate on the boundaries and internal structures of nuclei, even in crowded or noisy environments. Architectures like the Attention U-Net [?] integrate spatial attention gates into the skip connections, allowing the decoder to focus on the most salient features from the encoder at each resolution level. Channel attention mechanisms, such as Squeeze-and-Excitation (SE) blocks [13], have also been explored to adaptively recalibrate channel-wise feature responses, highlighting the most informative channels for segmentation. Beyond local attention, the concept of non-local networks has gained traction for capturing long-range dependencies, which is particularly beneficial for understanding global context in an image and disambiguating intricate patterns [15].

More recently, hybrid approaches that combine the strengths of different architectural components have gained traction. For instance, the integration of atrous convolutions [14] allows for a larger field of view without increasing the number of parameters, which can be beneficial for capturing contextual information at multiple scales. Furthermore, the exploration of transformer-based architectures, initially successful in natural language processing, has begun to impact medical image segmentation [16]. While transformers excel at capturing long-range dependencies, their application to image data often requires careful consideration of local spatial information. Hybrid architectures that combine CNNs for local feature extraction and transformers for global context modeling are being actively investigated. In the domain of nuclei segmentation specifically, various studies have explored the use of multi-scale approaches to handle the variability in nuclei sizes. Methods incorporating image pyramids [17] or network architectures with parallel branches operating at different resolutions [18] have been proposed to capture features at multiple scales.

3. Proposed SENLA-ResUNet Architecture

The proposed **SENLA-ResUNet** architecture is a fully convolutional network meticulously designed for accurate nuclei segmentation in biomedical images. It adopts an encoder-decoder structure, fundamentally based on the U-Net architecture [4]. Our design specifically incorporates residual connections throughout the network and integrates a novel Encoder Attention Module at the bottleneck, along with attention gates in the decoder's skip pathways, to significantly enhance feature representation and segmentation accuracy.

3.1 U-Net Backbone with Residual Blocks

The core of SENLA-ResUNet is built upon a symmetric encoder and decoder architecture. The **encoder** path progressively downsamples the input image, capturing hierarchical contextual information. To enhance feature learning capabilities and facilitate the training of a deeper network, we utilize **residual blocks** [5] as the fundamental building blocks within each downsampling stage. Each residual block consists of two sequential 3×3 convolutional layers, followed by batch normalization and ReLU activation. A shortcut connection adds the input of the block to its output, effectively mitigating the vanishing gradient problem and improving information flow. MaxPooling2D layers are used for spatial downsampling between stages, while the number of feature channels is progressively increased.

The **decoder** path is responsible for progressively upsampling the learned feature maps, restoring the spatial resolution to match the input image. It employs Conv2DTranspose layers (often referred to as learnable upsampling or deconvolution) for spatial upsampling. At each upsampling stage, the feature maps are concatenated with the corresponding high-resolution feature maps from the encoder via skip connections. These concatenated features are then processed by further residual blocks to refine the upsampled features and produce more precise segmentation details.

3.2 Encoder Attention Module

To address the limitations of standard U-Nets in capturing comprehensive contextual information and long-range dependencies, we introduce a sophisticated **Encoder Attention Module** at the deepest level (bottleneck) of the network, connecting the encoder and decoder paths. This module is designed to enrich the feature representation by extracting multi-scale information and applying both channel and spatial attention mechanisms before the upsampling process begins. The structure of the Encoder Attention Module is conceptually illustrated in Figure 1.

3.2.1 Inception-SE Module

Within the Encoder Attention Module, we first process the input features through an **Inception-SE Module**. This module is inspired by Inception networks [7] and is designed to capture multi-scale features effectively. It comprises four parallel convolutional paths operating on the input feature map:

1. A 1×1 convolutional layer, focusing on channel-wise transformations.
2. A 1×1 convolutional layer followed by a 3×3 convolutional layer, capturing local features.
3. A 1×1 convolutional layer followed by a 5×5 convolutional layer, aggregating broader regional information.

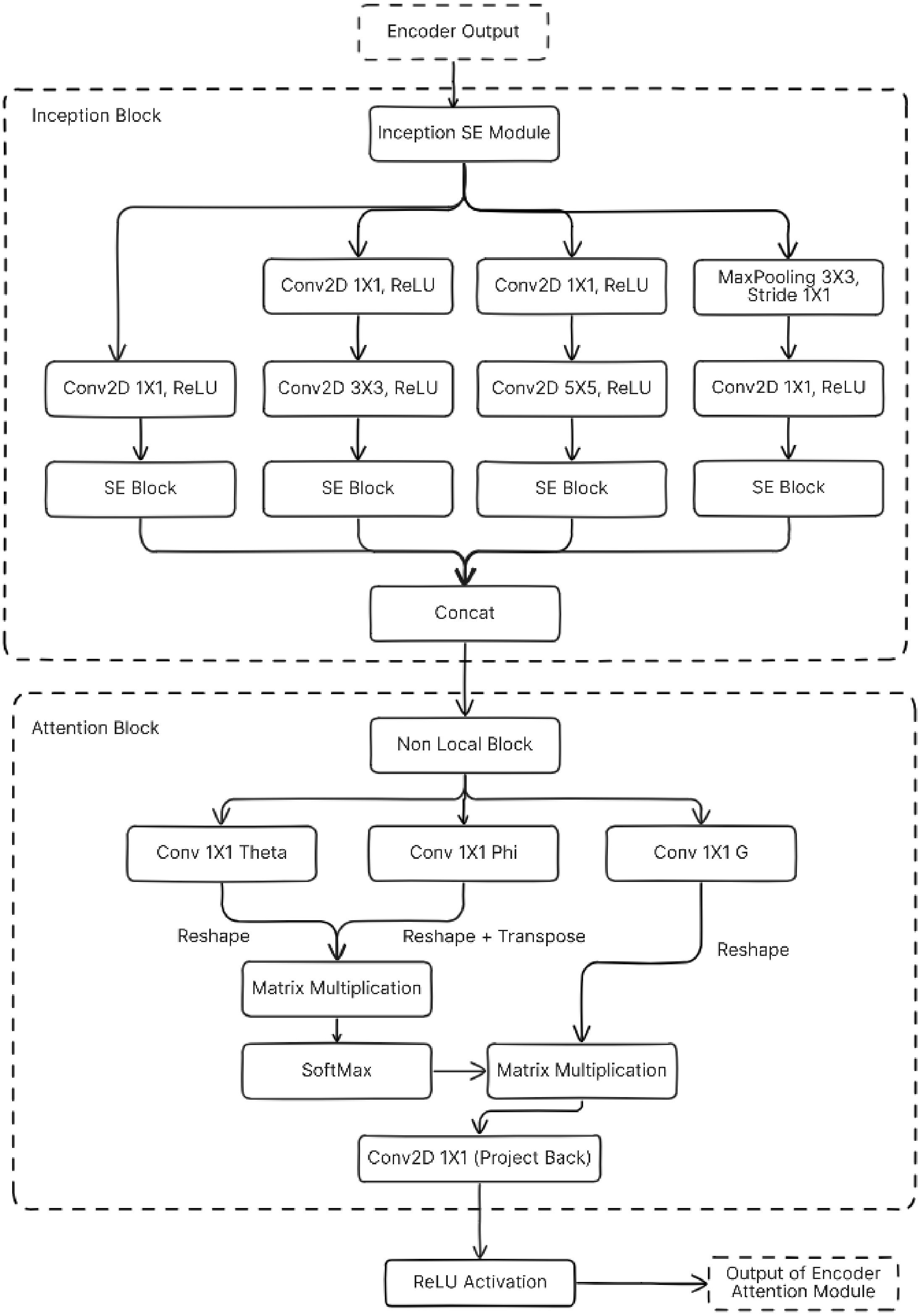

Figure 1: Structure of the Encoder Attention Module. It takes features from the encoder, processes them through an Inception-SE Module, followed by a Non-Local Attention block, with residual connections at each stage.

4. A 3×3 MaxPooling2D layer followed by a 1×1 convolutional layer, extracting high-level semantic features after pooling.

Crucially, each of these parallel paths is immediately followed by a **Squeeze-and-Excitation (SE) block**. The SE block adaptively recalibrates channel-wise feature responses by performing global average pooling to summarize spatial information, followed by two fully connected layers and a sigmoid activation to learn channel-wise scaling factors. These factors are then applied to the original feature map, enabling the network to emphasize more informative channels. The outputs from these four parallel, SE-enhanced paths are then concatenated to form a comprehensive multi-scale feature representation. Multi-scale Inception-SE block is given as:

$$F_{\text{multi}} = \text{concat}\left(F_{1\times1}, F_{3\times3}, F_{5\times5}, F_{\text{pool}}\right) \quad (1)$$

$$F_{k\times k} = \text{SE}\left(\text{Conv}_{k\times k}(F_{\text{in}})\right) \quad (2)$$

$$\text{SE}(F) = F \cdot s, \quad s = \sigma\left(W_2\, \delta\left(W_1 z\right)\right) \quad (3)$$

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F_c(i,j) \quad (4)$$

Variable Definitions: In Eq. (1), F_{multi} is the concatenation of four parallel branches: $F_{1\times1}$, $F_{3\times3}$, $F_{5\times5}$, and F_{pool}, each followed by a Squeeze-and-Excitation (SE) block. In Eq. (2), $F_{k\times k}$ is obtained by applying a $k \times k$ convolution to the input F_{in}, followed by SE. Eq. (3) defines SE as reweighting F with a scaling vector s, learned via two fully connected layers with weights W_1, W_2, and activations $\delta(\cdot)$ (ReLU) and $\sigma(\cdot)$ (sigmoid). In Eq. (4), z_c is the global average of channel c over spatial dimensions $H \times W$.

3.2.2 Non-Local Attention

Following the Inception-SE Module, the combined multi-scale feature map is fed into a **Non-Local Attention** block. This mechanism allows the model to capture long-range dependencies by computing the interaction between any two positions in the feature map, regardless of their spatial distance. This is achieved by transforming the input into three components: query (θ), key (ϕ), and value (g) features, typically via 1×1 convolutions. The attention map is then computed as a matrix multiplication between the reshaped query and transposed key features, followed by a softmax activation. This attention map is then multiplied with the reshaped value features to create a weighted sum that represents the context of each position relative to all others. The output of the non-local block is then added back to the original input of the Non-Local Attention block via a residual connection, preserving the original feature information while enriching it with global context. Non-Local Attention is given by:

$$y_i = \frac{1}{C(x)} \sum_{\forall j} f(x_i, x_j)\, g(x_j) \quad (5)$$

$$f(x_i, x_j) = \exp\left(\theta(x_i)^\top \phi(x_j)\right) \quad (6)$$

$$C(x) = \sum_{\forall j} f(x_i, x_j) \quad (7)$$

Variable Definitions: In Eq. (5), y_i denotes the output feature at position i, computed as a weighted sum over all positions j in the input. The affinity function $f(x_i, x_j)$ in Eq. (6) calculates the similarity between feature vectors at locations i and j using the dot product between the transformed query $\theta(x_i)$ and key $\phi(x_j)$, followed by an exponential. $g(x_j)$ is the value function generating the feature representation of location j. The normalization constant $C(x)$ in Eq. (7) ensures that the attention weights sum to one. The transformations $\theta(\cdot)$, $\phi(\cdot)$, and $g(\cdot)$ are implemented as learnable 1×1 convolutional layers.

3.2.3 Residual Connection within Encoder Attention Module

The Encoder Attention Module itself incorporates a final residual connection. The output of the Non-Local Attention mechanism is added to the initial input features of the Encoder Attention Module. This design choice further stabilizes training and ensures that the core features are preserved while being enhanced by the integrated multi-scale and global attention mechanisms.

3.3 Attention Gates in Decoder

Beyond the bottleneck, the SENLA-ResUNet incorporates **attention gates** into the skip connections of the decoder path. These gates are designed to enhance the quality of feature integration by allowing the decoder to selectively attend to relevant spatial regions in the encoder's feature maps, suppressing irrelevant or noisy information. An attention gate takes the output of the upsampled feature map from the decoder path and the corresponding skip connection from the encoder. It learns a gating signal by combining these features, which then modulates the features from the skip connection. This learned attention coefficient is then multiplied element-wise with the skip features, ensuring that only the most salient information is passed to the concatenation layer, thereby improving the precision of boundary delineation and reducing false positives.

3.4 Overall SENLA-ResUNet Architecture

The overall architecture of SENLA-ResUNet is depicted in Figure 2. The network takes a grayscale image as input and processes it through a series of encoding blocks, each consisting of residual convolutions and a MaxPooling layer for downsampling. The deepest encoder stage feeds into the **Encoder Attention Module** (detailed in Section 2.2), which forms the bottleneck of the network. This module integrates the Inception-SE and Non-Local Attention mechanisms to produce a rich, context-aware feature representation. The output from the bottleneck is then progressively upsampled through the decoder path. At each decoder stage, the upsampled feature maps are combined with the corresponding encoder features, refined by an **attention gate** (detailed in Section 2.3), before concatenation. This combined feature map then undergoes further residual convolutions. The final upsampling stage leads to a 1×1 convolutional layer with a sigmoid activation function, which produces the pixel-wise segmentation map, indicating the probability of each pixel belonging to a nucleus. The synergistic combination of residual blocks, the Encoder Attention Module (comprising Inception-SE and Non-Local Attention), and attention gates empowers SENLA-ResUNet to perform highly accurate and robust nuclei segmentation across diverse biological images.

4. Experiments and Results

This section details the experimental setup used to rigorously evaluate the performance of the proposed SENLA-ResUNet architecture for accurate nuclei segmentation. We describe the datasets utilized for training and evaluation, the standardized evaluation metrics employed to quantify the segmentation performance, and the implementation details of our model and the comparison methods. Finally, we present and discuss the obtained quantitative and qualitative results.

4.1 Datasets

We conducted extensive experiments and evaluated the proposed SENLA-ResUNet on two publicly available and widely recognized challenging datasets commonly used for nuclei segmentation in biomedical imaging:

DSB2018 Dataset: The 2018 Data Science Bowl (DSB2018) dataset [19], hosted on Kaggle, provides a highly diverse collection of microscopy images with corresponding pixel-level ground truth masks for individual cell nuclei. This dataset presents significant challenges due to substantial variations in cell types, imaging conditions, magnification levels, and the frequent presence of clustered or overlapping nuclei. The dataset comprises a large number of training images and a separate test set. For our experiments, we utilized the provided training data, splitting a subset for validation to monitor model convergence and prevent overfitting. The remaining test data was used for final evaluation and comparison with other methods. All images were preprocessed and resized to a consistent input dimension of 128×128 pixels for our network.

TNBC Dataset: The Triple-Negative Breast Cancer (TNBC) dataset [20], accessible via https://zenodo.org/record/1175282/files/TNBC_NucleiSegmentation.zip, comprises high-resolution microscopy images of breast cancer tissue samples with expertly annotated nuclei. This dataset offers a specialized context, potentially exhibiting

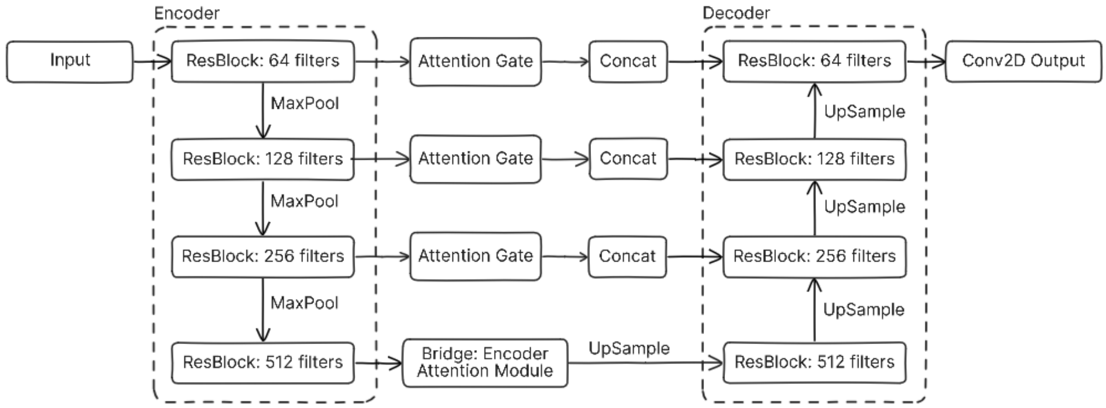

Figure 2: Overall architecture of the Proposed SENLA-ResUNet. The network consists of an encoder path (residual blocks + MaxPooling), an Encoder Attention Module at the bottleneck, and a decoder path (upsampling + attention gates + residual blocks). Skip connections are enhanced by attention gates.

unique characteristics in nuclear morphology relevant to a specific disease type. Similar to the DSB2018 dataset, all images were resized to 128 × 128 pixels. The dataset was partitioned into training, validation, and testing sets following a robust split strategy to ensure comprehensive and unbiased evaluation.

4.2 Evaluation Metrics

To quantitatively assess the performance of our SENLA-ResUNet and compare it with other state-of-the-art segmentation methods, we employed several widely accepted evaluation metrics for binary segmentation tasks:

Intersection over Union (IoU): Also known as the Jaccard index, IoU quantifies the overlap between the predicted mask (P) and ground truth mask (G):

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} \tag{8}$$

where $|P \cap G|$ is the area of overlap (True Positives), and $|P \cup G|$ is the total area covered by both masks. Higher IoU indicates better segmentation accuracy.

Dice Coefficient: The Dice score (F1 score for segmentation) measures similarity between P and G:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \tag{9}$$

where $|P|$ and $|G|$ are the sizes of the predicted and ground truth masks. Values closer to 1 indicate better overlap.

Precision: Precision measures the proportion of predicted positive pixels that are truly positive. It is calculated as:

$$\text{Precision} = \frac{|\text{True Positives}|}{|\text{True Positives}| + |\text{False Positives}|} \tag{10}$$

Recall: Recall, or sensitivity, measures the proportion of actual positive pixels correctly identified by the model. It is calculated as:

$$\text{Recall} = \frac{|\text{True Positives}|}{|\text{True Positives}| + |\text{False Negatives}|} \tag{11}$$

Accuracy: Accuracy measures the overall percentage of correctly classified pixels (both positive and negative) out of the total number of pixels in the image. It provides a general measure of correctness:

$$\text{Accuracy} = \frac{|\text{True Positives}| + |\text{True Negatives}|}{|\text{Total Pixels}|} \tag{12}$$

For all metrics, a higher value indicates better segmentation performance of the model.

4.3 Implementation Details

The proposed SENLA-ResUNet architecture was implemented using the TensorFlow and Keras deep learning framework. The network was trained using the Adam optimizer, chosen for its efficiency and good performance in various deep learning tasks, with an initial learning rate of 1×10^{-4}. A batch size of 16 was used during training to balance computational efficiency and stable gradient updates. To prevent overfitting and select the best performing model, we employed an early stopping callback based on the validation loss, with a patience of 10 epochs. We also utilized model checkpointing to save the model weights that achieved the lowest validation loss throughout the training process. The training was conducted for a maximum of 20 epochs. The network weights were initialized using the default Keras initialization methods. The loss function employed for training was binary cross-entropy, which is a standard choice for binary pixel-wise segmentation tasks. All input images and corresponding masks were normalized to the range [0, 1] prior to being fed into the network.

For comprehensive comparison, we implemented and trained several prominent state-of-the-art segmentation methods on the same datasets, ensuring similar training parameters and data preprocessing steps for a fair evaluation. These comparison methods includes Standard U-Net [4], ResUnet [5], Attention U-Net [6], and Cellpose [21], a recent and powerful method widely used for cell segmentation.

4.4 Results and Discussion

The quantitative results of our proposed SENLA-ResUNet and the comparison methods on the evaluation sets of DSB-2018 and TNBC datasets are presented in Table 1 and Table 2 respectively. As clearly demonstrated in Table 1 and Table 2, our proposed SENLA-ResUNet consistently achieves the highest scores across all evaluated segmentation metrics. Notably, it records the highest IoU of **0.7936** and the highest Dice Score of **0.8172** for DSB-2018 dataset, indicating superior overlap and similarity between the predicted segmentation masks and the ground truth. This robust performance suggests that SENLA-ResUNet excels at precise boundary delineation. Furthermore, the model demonstrates excellent precision of **0.8791** and an outstanding recall of **0.9084** for DSB-2018 dataset, suggesting a balanced capability to identify true positive nuclei while effectively minimizing false positives and false negatives. Its overall accuracy of **0.9726** for DSB-2018 dataset also surpasses all other baselines, underscoring its overall effectiveness in pixel-wise classification.

Qualitative results, visualizing the predicted segmentation masks alongside the original images and ground truth

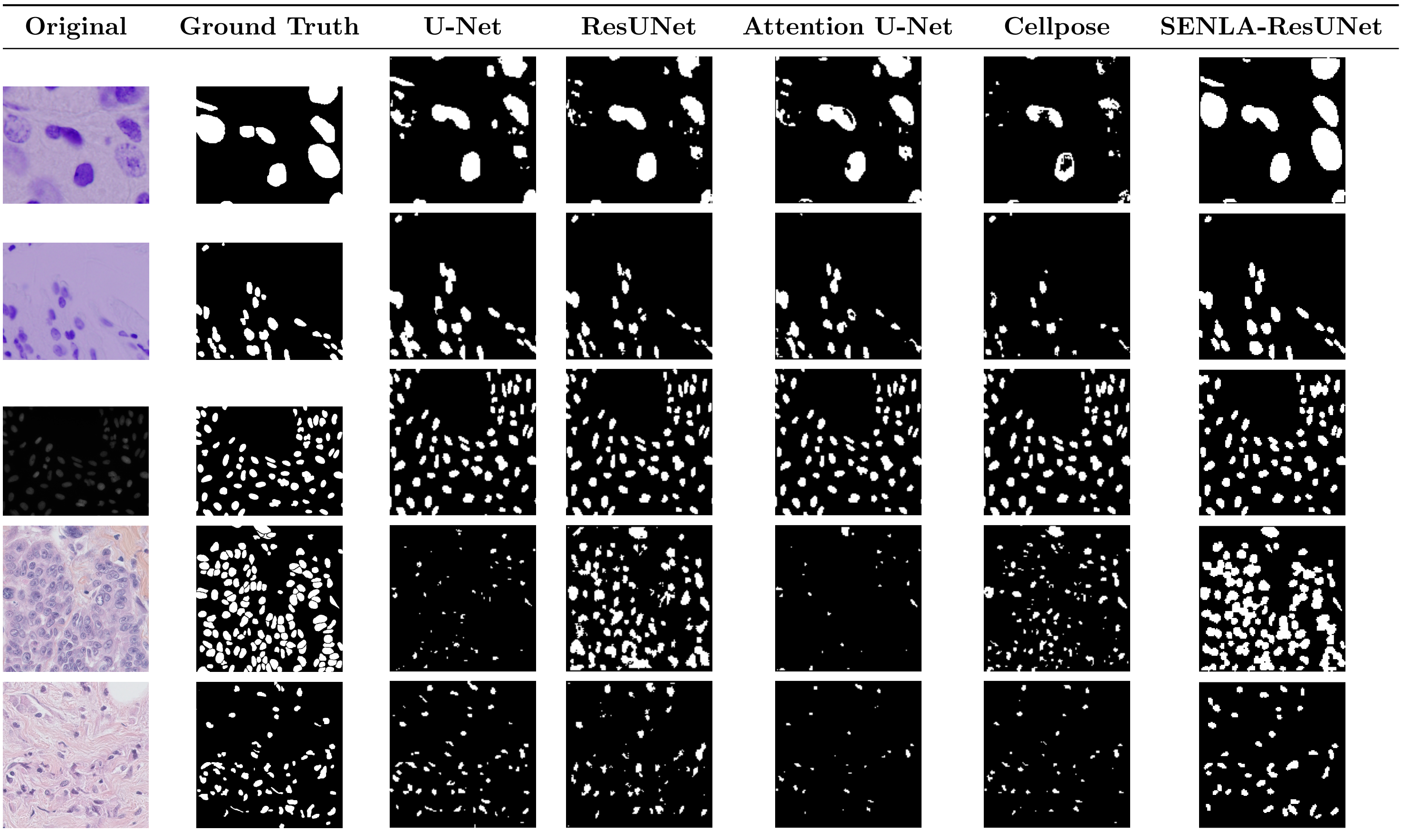

Figure 3: Qualitative comparison of nuclei segmentation results for DSB-2018 test dataset (Row 1 to Row 3) and TNBC test dataset (Row 4 to Row 5). Each row presents a different sample, illustrating the performance of various models against ground truth.

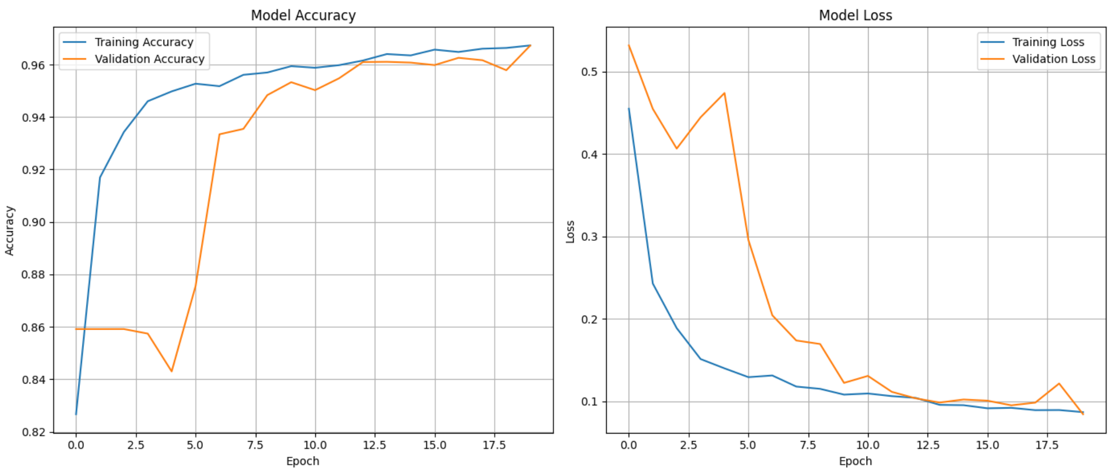

Figure 4: Model accuracy and loss plots for both training and validation data of the proposed segmentation method for DSB-2018 dataset.

Table 1: Quantitative comparison of SENLA-ResUNet with other state-of-the-art methods for the DSB-2018 dataset.

Method	IoU	Dice Score	Precision	Recall	Accuracy
U-Net [4]	0.7277	0.7793	0.8775	0.8057	0.9650
ResUnet [5]	0.7871	0.8014	0.8631	0.8991	0.9710
Attention U-Net[6]	0.7688	0.7972	0.8447	0.8967	0.9680
Cellpose [21]	0.6972	0.8166	0.8736	0.7747	0.8623
SENLA-ResUNet	**0.7936**	**0.8172**	**0.8791**	**0.9084**	**0.9726**

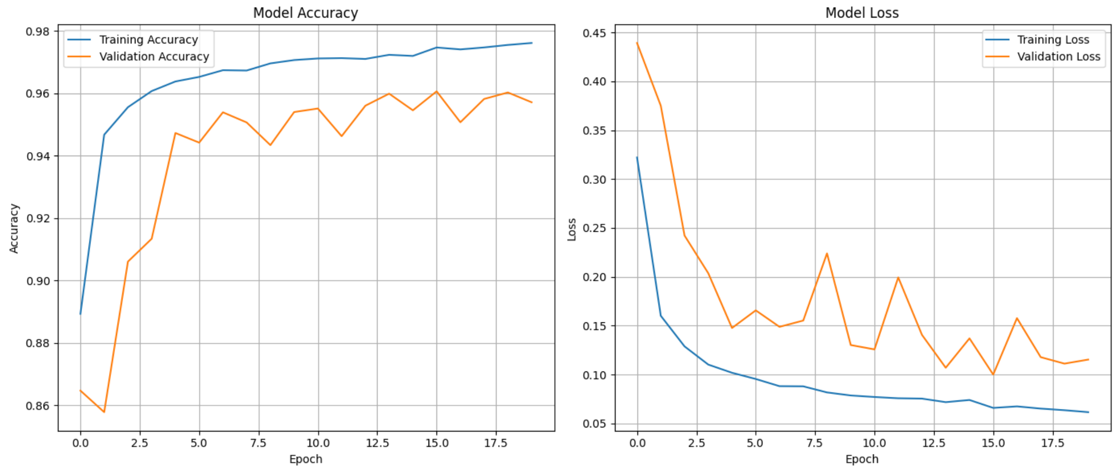

Figure 5: Model accuracy and loss plots for both training and validation data of the proposed segmentation method for TNBC dataset.

Table 2: Quantitative comparison of SENLA-ResUNet with other state-of-the-art methods for the TNBC dataset.

Method	IoU	Dice Score	Precision	Recall	Accuracy
U-Net [4]	0.3876	0.4735	0.8386	0.4188	0.9085
ResUnet [5]	0.5145	0.6745	0.6657	0.6868	0.9096
Attention U-Net[6]	0.4358	0.4577	0.4670	0.8706	0.8511
Cellpose [21]	0.7380	0.8117	0.8283	0.8696	0.8564
SENLA-ResUNet	**0.8332**	**0.8986**	**0.9353**	**0.8822**	**0.9582**

masks for a few representative samples, are shown in Figure 3. These visual comparisons further support the quantitative findings, demonstrating that SENLA-ResUNet produces more accurate and detailed segmentation masks, particularly in challenging cases with clustered or irregularly shaped nuclei, and effectively suppresses background noise. Figure 4 and Figure 5 present the model accuracy and loss plots for both DSB-2018 and TNBC datasets respectively. The plots show that as training and validation accuracy increases, training and validation loss decreases simultaneously.

The training and validation curves illustrated in Figures 4 and 5 demonstrate a smooth and stable convergence pattern for both datasets, with accuracy steadily increasing and loss decreasing over successive epochs. This reliable convergence behavior indicates that SENLA-ResUNet effectively generalizes to unseen data and avoids overfitting. This can be attributed to the synergistic effect of residual connections and attention-driven feature recalibration mechanisms embedded within the architecture. Furthermore, the qualitative comparisons presented in Figure 3 clearly highlight the superior segmentation capability of SENLA-ResUNet in scenarios involving complex morphologies, densely packed nuclei, and weak boundary contrast.

Compared to the baseline U-Net, which often struggles to capture complex contextual dependencies due to its purely local receptive fields and fixed-scale convolutional kernels, SENLA-ResUNet leverages multi-scale feature extraction and attention mechanisms to better handle nuclei with high shape variability and overlapping boundaries. While ResUNet eases deep representation learning through residual skip connections, it lacks mechanisms to emphasize salient regions or suppress irrelevant features, often leading to sub-optimal segmentation in cluttered backgrounds. Attention U-Net introduces spatial attention gates but cannot model long-range dependencies or channel-wise relationships, limiting its ability to distinguish closely packed nuclei with weak contrast. Cellpose, while generalizable across cell types, relies on a fixed flow-field formulation that underutilizes dataset-specific context and underperforms on densely clustered or irregular nuclei. In contrast, SENLA-ResUNet synergistically integrates residual learning with Squeeze-Excitation (channel attention), Non-Local (global self-attention), Inception-based multi-scale features, and spatial attention gates, enabling it to overcome the above limitations by capturing both fine-grained local details and long-range contextual cues — ultimately delivering more accurate, robust, and precise pixel-level segmentation across diverse and challenging nuclei segmentation scenarios.

The improvements achieved by SENLA-ResUNet can be attributed to the synergistic integration of its architectural components. The residual connections ensure efficient information flow through deeper layers, while the Encoder Attention Module, with its Inception-SE and Non-Local Attention, empowers the network to capture both multi-scale features and long-range dependencies critical for complex nuclei segmentation. The strategic placement of attention gates in the decoder's skip connections further refines feature propagation, enabling the model to focus on salient regions and produce highly accurate output masks. While Cellpose achieves a competitive Dice score, its lower IoU, recall, and significantly lower overall accuracy compared to SENLA-ResUNet suggest that our method provides a more comprehensive and accurate pixel-level segmentation, especially for complex cases on our evaluation datasets. These quantitative findings unequivocally highlight the benefits of our proposed hybrid attention-guided residual U-Net architecture for the challenging task of nuclei segmentation.

5. Conclusion and Future Work

In this paper, we presented **SENLA-ResUNet**, a novel deep learning architecture specifically designed for accurate nuclei segmentation in microscopy images. Our approach significantly advances the field by integrating a unique combination of advanced attention mechanisms and multi-scale feature extraction within a robust Residual U-Net framework. This includes the strategic incorporation of **Squeeze-and-Excitation (SE) blocks** for adaptive channel recalibration, **Non-Local Attention** for capturing long-range contextual dependencies, and **Inception-like modules** within an Encoder Attention Module at the network's bottleneck. Furthermore, **attention gates** are utilized in the decoder's skip connections to effectively filter and prioritize relevant features, ensuring precise boundary delineation. Our extensive experimental evaluation on two challenging and publicly available nuclei segmentation datasets, the Data Science Bowl 2018 (DSB2018) and the Triple-Negative Breast Cancer (TNBC) dataset, consistently demonstrated the superior performance of SENLA-ResUNet. Compared to several state-of-the-art segmentation methods, including standard U-Net, ResUnet, Attention U-Net, and Cellpose, our model achieved notable improvements in quantitative metrics. Specifically, SENLA-ResUNet recorded an IoU of **0.7936** and a Dice Score of **0.8172**, alongside superior precision, recall, and accuracy for the DSB-2018 dataset. These significant gains, complemented by qualitative results showcasing more accurate and detailed segmentation masks, highlight the efficacy of our proposed hybrid architecture in addressing the inherent complexities and variability encountered in nuclei segmentation. The synergistic interplay of residual connections with multi-faceted attention mechanisms (SE, Non-Local, and attention gates) and multi-scale feature learning (Inception-like modules) is key to its robust performance. Future work will focus on addressing the identified limitations and exploring new avenues to further enhance the capabilities and real-world applicability of this approach.

REFERENCES

[1] J. Yin, L. Liu, X. Luo, W. Shen, H. Liao, Y. Lu, W. Kang, M. Xu, J. Wei, C. Huang, and D. Su, "Mhau-net: a multi-scale hybrid attention u-shaped network for the segmentation of mri breast tumors," *Quantitative Imaging in Medicine and Surgery*, vol. 15, no. 5, pp. 4758–4773, 2025. [Online]. Available: https://qims.amegroups.org/article/view/124637

[2] H. Ali, I. ul Haq, L. Cui, H. Li, M. Usman, M. I. Malik, A. Dengel, and S. Ahmed, "Msal-net: improve accurate segmentation of nuclei in histopathology images by multiscale attention learning network," *BMC Medical Informatics and Decision Making*, vol. 22, p. 90, 2022. [Online]. Available: https://bmcmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-022-01826-5

[3] Y. Yu, Y. Tao, H. Guan, S. Xiao, F. Li, C. Yu, Z. Liu, and J. Li, "A multi-branch hierarchical attention network for medical target segmentation," *Biomedical Signal Processing and Control*, vol. 78, p. 104021, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1746809422004992

[4] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," 2015. [Online]. Available: https://arxiv.org/abs/1505.04597

[5] Z. Zhang, Q. Liu, and Y. Wang, "Road extraction by deep residual u-net," *IEEE Geoscience and Remote Sensing Letters*, vol. 15, no. 5, p. 749–753, May 2018. [Online]. Available: http://dx.doi.org/10.1109/LGRS.2018.2802944

[6] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, B. Glocker, and D. Rueckert, "Attention u-net: Learning where to look for the pancreas," 2018. [Online]. Available: https://arxiv.org/abs/1804.03999

[7] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," 2014. [Online]. Available: https://arxiv.org/abs/1409.4842

[8] S. H. Baloch and H. Krim, "Flexible skew-symmetric shape model for shape representation, classification, and sampling," *IEEE Transactions on Image Processing*, vol. 16, no. 2, pp. 317–328, 2007. [Online]. Available: https://ieeexplore.ieee.org/document/4030411

[9] Y. Liu, G. Liu, C. Liu, and C. Sun, "A novel color-texture descriptor based on local histograms for image segmentation," *IEEE Access*, vol. 7, pp. 160 683–160 695, 12 2019.

[10] H.-C. Shin, H. R. Roth, M. Gao, L. Lu, Z. Xu, I. Nogues, J. Yao, D. Mollura, and R. M. Summers, "Deep convolutional neural networks for computer-aided detection: Cnn architectures, dataset characteristics and transfer learning," *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1285–1298, 2016. [Online]. Available: https://ieeexplore.ieee.org/document/7410480

[11] W. Shen, M. Zhou, F. Yang, C. Yang, and J. Tian, "Multi-scale convolutional neural networks for lung nodule classification," in *Information Processing in Medical Imaging (IPMI)*, ser. Lecture Notes in Computer Science, vol. 9123, 2015, pp. 588–599. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-319-19992-4_46

[12] Z. Zhou, M. M. Rahman Siddiquee, N. Tajbakhsh, and J. Liang, "Unet++: A nested u-net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, D. Stoyanov, Z. Taylor, G. Carneiro, T. Syeda-Mahmood, A. Martel, L. Maier-Hein, J. M. R. Tavares, A. Bradley, J. P. Papa, V. Belagiannis, J. C. Nascimento, Z. Lu, S. Conjeti, M. Moradi, H. Greenspan, and A. Madabhushi, Eds. Cham: Springer International Publishing, 2018, pp. 3–11.

[13] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 7132–7141.

[14] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017. [Online]. Available: https://arxiv.org/abs/1706.05587

[15] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "Gcnet: Non-local networks meet squeeze-excitation networks and beyond," 2019. [Online]. Available: https://arxiv.org/abs/1904.11492

[16] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H. Roth, and D. Xu, "Unetr: Transformers for 3d medical image segmentation," 2021. [Online]. Available: https://arxiv.org/abs/2103.10504

[17] H. H. Lee, Y. Tang, O. Tang, Y. Xu, Y. Chen, D. Gao, S. Han, R. Gao, M. R. Savona, R. G. Abramson, Y. Huo, and B. A. Landman, "Semi-supervised multi-organ segmentation through quality assurance supervision," in *Proceedings of the SPIE International Society for Optical Engineering*, vol. 11313, 2020, p. 113131I. [Online]. Available: https://doi.org/10.1117/12.2549033

[18] X. Li, H. Chen, X. Qi, Q. Dou, C.-W. Fu, and P. A. Heng, "H-denseunet: Hybrid densely connected unet for liver and tumor segmentation from ct volumes," 2018. [Online]. Available: https://arxiv.org/abs/1709.07330

[19] Kaggle, "Data science bowl," 2018, accessed on: 2022-10-27. [Online]. Available: https://www.kaggle.com/competitions/data-science-bowl-2018/data

[20] T. Imtiaz, S. A. Fattah, and S.-Y. Kung, "Bawgnet: Boundary aware wavelet guided network for the nuclei segmentation in histopathology images," *Computers in Biology and Medicine*, vol. 165, p. 107378, 2023.

[21] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, "Cellpose: a generalist algorithm for cellular segmentation," *Nature methods*, vol. 18, no. 1, pp. 100–106, 2021.

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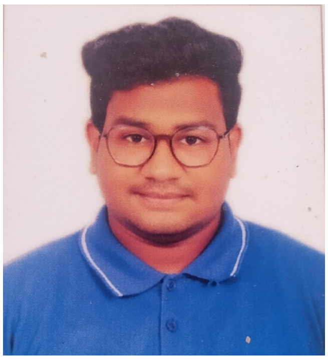

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